

PRD — Agent-Generated Preference Data for DPO Fine-Tuning

Goals, scope, stakeholders, success metrics, timeline, and risks for the project.

Project: Agent-Generated Preference Data for DPO Fine-Tuning of a Small Language Model
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Budget constraint: Every component of this project is designed to run at \$0 cost — free-tier compute (Google Colab T4), free/open-weight models, and free public datasets only.

1. Purpose

Define what is being built, for whom, and why — independent of implementation detail. This PRD treats the deliverable as a **research artifact**: a trained model, an evaluation report, and a reusable pipeline, rather than a consumer product.

2. Goals

- Produce a small language model (1.5-3B params) measurably aligned via DPO on agent-generated preference data, at \$0 infrastructure cost.
- Produce a rigorous, reproducible **ablation study** comparing agent-generated vs. human-labeled preference data of equal size.
- Produce a reusable, documented pipeline (agent debate → preference dataset → DPO training → evaluation) re-runnable on a different base model or domain in under a day.
- Produce a written report structured for either an RVCE publication track or a portfolio case study, including an explainability component.

3. Non-Goals

- Not attempting to beat frontier-model alignment quality — the target is a small model, not a production chat assistant.
- Not doing full RLHF/PPO — DPO only, by design, for cost and stability reasons.
- Not collecting any real human preference labels — human-labeled data is used only as an already-published reference set (HH-RLHF), never newly annotated.
- Not deploying the model as a public-facing service; the deliverable is the model weights, code, and report.

4. Stakeholders and Users

Stakeholder	Who	What they care about
Primary author	Dhruv	Designs, builds, and evaluates the full pipeline
Academic evaluator	RVCE faculty / SIH-style jury	Novelty, rigor, and correctness of the ablation
Future self / portfolio readers	Recruiters, grad-school reviewers	Clarity of the ML contribution, not agent-orchestration novelty

5. Success Metrics

Metric	Target	How measured
Win-rate uplift	DPO-tuned model beats its own untuned base in $\geq 60\%$ of judged head-to-head comparisons on a held-out prompt set	LLM-judge win-rate

Metric	Target	How measured
Gap to human-labeled baseline	Agent-data model's win-rate is within 10 percentage points of the human-data-trained model	Same judge harness, same prompts
Benchmark movement	Measurable, reported change on TruthfulQA / HaluEval subset scores, with confidence intervals	Automated scoring scripts
Reproducibility	End-to-end pipeline re-runs from a fresh Colab session in under 3 hours on a free T4	Wall-clock timing
Report quality	Ablation table + SHAP feature-importance figure + written discussion included in final report	Manual review

6. Scope

In scope

- 1-3B parameter base model, LoRA/QLoRA fine-tuning only
- DPO training via TRL's DPOTrainer (optionally Unsloth-accelerated)
- Agent debate + critic pipeline for preference-pair generation
- Optional small reward model + SHAP explainability layer
- Evaluation on 1-2 public benchmarks plus a custom held-out prompt set

Out of scope

- Multi-turn conversational memory / long-context evaluation
- Multi-modal (image/audio) capabilities
- Any paid API usage beyond free tiers
- Production serving, latency optimization, or a user-facing app

7. Milestones and Timeline

Milestone	Deliverable
Week 1	Research finalized, model chosen, environment set up, baseline (untuned) model evaluated on benchmarks
Week 2	Agent debate + critic pipeline built and generating preference pairs; dataset schema finalized and validated
Week 3	First DPO training run (LoRA) on agent-generated data; smoke-test evaluation
Week 4	Second DPO run on equal-sized human-labeled (HH-RLHF) subset for comparison
Week 5	Full evaluation harness run on both models; ablation table assembled
Week 6	Reward model + SHAP explainability layer; write-up and report finalized

8. Risks and Mitigations

Risk	Likelihood/Note	Mitigation
Free-tier API rate limits stall data generation	Medium	Fall back to a local Ollama model for the agent debate layer — fully free, no rate limits
Colab free-tier GPU session timeouts mid-training	Medium	Checkpoint every N steps to Google Drive / Hugging Face Hub; keep runs under ~2 hours per session
Agent-generated preferences are systematically biased (e.g. prefer longer answers)	High — an interesting finding, not just a risk	Explicitly measure and report this bias as part of the ablation rather than trying to eliminate it
DPO training destabilizes the small model (catastrophic forgetting)	Low-Medium	Conservative learning rate (5e-6 to 5e-5), low LoRA rank (8-16), monitor reward margin / KL divergence

References & Links

The PRD references the same core techniques and free-tier resources detailed fully in the Research Report; links below are the anchors most relevant to scope and feasibility decisions in this document.

Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C.D., Finn, C. (2023). **Direct Preference Optimization: Your Language Model is Secretly a Reward Model**. <https://arxiv.org/abs/2305.18290> — the core DPO algorithm used for training

Bai, Y. et al. (2022). **Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback**. <https://arxiv.org/abs/2204.05862> — source paper for the HH-RLHF human-labeled reference dataset

Lin, S., Hilton, J., Evans, O. (2021). **TruthfulQA: Measuring How Models Mimic Human Falsehoods**. <https://arxiv.org/abs/2109.07958> — evaluation benchmark used in this project

Li, J. et al. (2023). **HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models**. <https://arxiv.org/abs/2305.11747> — second evaluation benchmark used in this project

Google. **Google Colaboratory — Free Tier**. <https://colab.research.google.com> — free T4 GPU compute used for all training

Unsloth AI. **Unsloth — GitHub Repository & Docs**. <https://github.com/unslothai/unsloth> — free-tier-optimized LoRA/QLoRA/DPO training